

# BST Physics Curriculum Vol 11 Chapter 7 — Heegner Numbers + Cremona Elliptic Curves v0.4 SIGNATURE (refilled per Cal #104)

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## Vol 11 Chapter 7 — Heegner Numbers + Cremona Elliptic Curves

### Chapter motivation

Heegner-Stark theorem (Heegner 1952 + Stark 1967): imaginary quadratic field  $\mathbb{Q}(\sqrt{-d})$  has class number 1 iff  $d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$  — the 9 Heegner discriminants. Used by Heegner 1952 to give first proof of  $d = 163$  (using  $j$ -invariant near-integer  $e^{\pi\sqrt{163}} \approx 262537412640768744$  integer to  $10^{-12}$  precision); rigorized by Stark 1967.

Cremona elliptic curves database (Cremona 1992-current): tables of elliptic curves over  $\mathbb{Q}$  ordered by conductor; canonical labels (49a1, 27a1, 121a1, etc.); millions of curves with computed invariants (rank, torsion, discriminant, CM type, modular form, etc.).

BST connection: Cremona 49a1 is BST canonical elliptic curve  $Y^2 = X^3 - 945X - 10206$  with conductor  $49 = g^2$ , discriminant  $-343 = -g^3$ ,  $j$ -invariant  $-15^3 = -(N_c \cdot n_C)^3$ , complex multiplication by  $\mathbb{Q}(\sqrt{-g}) = \mathbb{Q}(\sqrt{-7})$ , rank = 2, torsion = 2 — ALL invariants BST primary. Per Vol 0 Ch 9 Strong-Uniqueness C9 (Stark small-primary subset anchoring): K75 + T2461 STRUCTURALLY VERIFIED at  $\dim_C = 5$  Friday —  $D_{IV}^5$  uniquely supports Cremona 49a1 anchoring at  $\{-3, -7, -11\}$  Stark subset of Heegner discriminants.

### Section 7.0b — Reader-grade 3-level pedagogy

Level 1: 9 Heegner discriminants  $\{1, 2, 3, 7, 11, 19, 43, 67, 163\}$  from Stark-Heegner 1952-1967; Cremona 49a1  $= Y^2 = X^3 - 945X - 10206$  BST canonical elliptic curve at BST primary  $g^2$  conductor; T2461 Strong-Uniqueness C9 STRUCTURALLY VERIFIED at  $\dim_C = 5$  (Friday) —  $D_{IV}^5$  uniquely supports.

Level 2 (graduate-mathematician): Stark-Heegner theorem: imaginary quadratic field  $Q(\sqrt{-d})$  has class number  $h(d) = 1$  iff  $d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$  (the 9 Heegner discriminants). Heegner 1952 first proof (using  $j$ -invariant near-integer phenomenon): for  $d = 163$ ,  $e^{\{\pi\sqrt{163}\}} \approx 262537412640768744$  (Ramanujan constant integer to  $10^{-12}$  precision); near-integer because  $e^{\{\pi\sqrt{d}\}} \approx -j((1+\sqrt{-d})/2) - 744 + (\text{small term})$  where  $j$  is the  $j$ -invariant of complex multiplication elliptic curve with CM by  $Q(\sqrt{-d})$ ; for  $d = 163$ ,  $j$  is integer  $-640320^3$ , so near-integer to  $\sim 10^{-12}$ . Stark 1967 rigorized + extended to negative-result for  $d \notin$  Heegner. Cremona Database of Elliptic Curves: tables of  $E/Q$  ordered by conductor with computed invariants (rank, torsion, discriminant, CM, modular form, Tamagawa numbers, etc.). BST canonical curve Cremona 49a1:  $Y^2 = X^3 - 945X - 10206$ ; conductor  $N = 49 = g^2$  (BST primary squared); discriminant  $\Delta = -343 = -g^3$  (BST primary cubed);  $j$ -invariant  $j = -15^3 = -(N_c \cdot n_C)^3 = -3375$  (BST primary product cubed); complex multiplication by  $Q(\sqrt{-g}) = Q(\sqrt{-7})$  (Stark-Heegner small subset); rank  $r(E) = 2 = \text{rank}$  (BST primary); torsion  $E(Q)_{\text{tors}} = \mathbb{Z}/2\mathbb{Z}$  (rank-2 Mordell-Weil); modular form  $f_E$  of weight 2 level 49. All invariants BST primary integer or BST primary product. Vol 0 Ch 9 Strong-Uniqueness C9 (Stark small-primary subset anchoring  $\{-3, -7, -11\}$ ):  $K75 + T2461$  STRUCTURALLY VERIFIED at  $\dim_C = 5$  Friday —  $D_{IV}^5$  uniquely supports Cremona 49a1 anchoring with the small-primary subset  $\{-3, -7, -11\}$  (Heegner discriminants 3, 7, 11) matching BST primary integers  $N_c, g, c_2$ . T1430 1/rank universality:  $L(E, 1)/\Omega(E) = 1/\text{rank}(E) = 1/2$  for Cremona 49a1 (BST 49a1 = “canonical elliptic curve for substrate”). Vol 0 Ch 9 C18  $D_{IV}^5$  Rigidity Principle (T2467 + T2468) inherits 49a1 anchoring via Bridge Object Family 1 (Heegner-trio  $K47 = \text{Cremona 49a1}$  at BST primary discriminant  $-g = -7$ ).

Level 3 (5th-grader accessible): Heegner-Stark theorem (1952-1967): there are exactly 9 imaginary quadratic fields  $Q(\sqrt{-d})$  with class number 1 — the 9 Heegner discriminants  $\{1, 2, 3, 7, 11, 19, 43, 67, 163\}$ . BST’s canonical elliptic curve is Cremona 49a1:  $Y^2 = X^3 - 945X - 10206$  with conductor  $49 = 7^2$  (BST integer  $g$  squared), discriminant  $-343 = -7^3$ ,  $j$ -invariant  $-15^3$  where  $15 = N_c \cdot n_C = 3 \cdot 5$ . All invariants are BST primary integers — making 49a1 the substrate-natural elliptic curve. T2461 (Friday) Strong-Uniqueness C9 STRUCTURALLY VERIFIED:  $D_{IV}^5$  uniquely supports this anchoring.

## Section 7.1 — Stark-Heegner Theorem (1952-1967)

Imaginary quadratic field  $Q(\sqrt{-d})$  class number  $h(d) = 1$  iff  $d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$ .

Heegner 1952 first proof via  $j$ -invariant near-integer phenomenon (Ramanujan constant  $e^{\{\pi\sqrt{163}\}}$  integer to  $10^{-12}$ ).

Stark 1967 rigorized + extended negative-result  $d \notin$  Heegner.

## Section 7.2 — Cremona Elliptic Curves Database

Cremona 1992-current: tables of E/Q ordered by conductor; canonical labels (49a1, 27a1, 121a1, ...); computed invariants (rank, torsion, discriminant, CM, modular form, Tamagawa).

## Section 7.3 — Cremona 49a1 BST Canonical Elliptic Curve

$$Y^2 = X^3 - 945X - 10206$$

- Conductor:  $N = 49 = g^2$  (BST primary squared)
- Discriminant:  $\Delta = -343 = -g^3$  (BST primary cubed)
- j-invariant:  $j = -15^3 = -(N_c \cdot n_C)^3 = -3375$
- CM:  $Q(\sqrt{-g}) = Q(\sqrt{-7})$  (Heegner discriminant 7)
- Rank:  $r(E) = 2 = \text{rank (BST primary)}$
- Torsion:  $E(Q)_{\text{tors}} = \mathbb{Z}/2\mathbb{Z}$

ALL invariants BST primary integer or BST primary product.

## Section 7.4 — T2461 Strong-Uniqueness C9 STRUCTURALLY VERIFIED at $\dim_C = 5$

Vol 0 Ch 9 Strong-Uniqueness C9 (Stark small-primary subset anchoring  $\{-3, -7, -11\}$ ): K75 + T2461 STRUCTURALLY VERIFIED at  $\dim_C = 5$  Friday.

$D_{IV}^5$  uniquely supports Cremona 49a1 anchoring with the small-primary subset  $\{-3, -7, -11\}$  matching BST primaries ( $N_c = 3, g = 7, c_2 = 11$ ).

## Section 7.5 — T1430 1/Rank Universality

$L(E, 1)/\Omega(E) = 1/\text{rank}(E) = 1/2$  for Cremona 49a1.

“Canonical elliptic curve for substrate” — all 7 Millennium problems + Four-Color Theorem reduce to 1/rank framework via Cremona 49a1 anchoring.

## Section 7.6 — Connection Vol 0 Ch 9 C18 $D_{IV}^5$ Rigidity Principle

Vol 0 Ch 9 C18  $D_{IV}^5$  Rigidity Principle (T2467 + T2468 Friday) inherits 49a1 anchoring via Bridge Object Family 1 (Heegner-trio K47 = Cremona 49a1 at BST primary discriminant  $-g = -7$ ).

Substrate-singleton via Rigidity Principle  $\leftrightarrow$  Cremona 49a1 substrate-canonical anchoring.

## Section 7.7 — Honest scope + Connection

- Stark-Heegner 1952-1967 standard [\[?\]](#)
- Cremona 49a1 all-invariants-BST-primary cross-link [\[?\]](#)

- T2461 Strong-Uniqueness C9 STRUCTURALLY VERIFIED at  $\dim_C = 5$  ?
- T1430 1/rank universality + 7 Millennium reduction
- Open scope: explicit Cremona-database substrate-cartography catalog (multi-month)

Connection: - Vol 0 Ch 9 Strong-Uniqueness C9 + C18 (Stark anchor + Rigidity)  
 - Vol 4 Ch 4 T1924  $t_{\text{cosmo}} = 47$  cross-link (Monster + 49a1 anchoring chain) -  
 Vol 11 Ch 8 Monster Moonshine + Supersingular Primes - T2467 + T2468  $D_{IV}^5$   
 Rigidity Principle (Casey-named #7)

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